

Macro Regime and Threshold Experiments Report

I created an experiment pipeline macro regime v2 experiment suite and spent the last few days reconciling those results against Rohit sir's consolidated plan PDF. I have added answers to the questions that you had asked along with my doubts for you.

Part A: Five regime dimensions

A1: fed_cycle (7 states → 4 states)

Question	Answered?	Answer
Does 4-state collapse give enough observations per state?	Partially	TIGHTENING 763 (40.1%), EASING 727 (38.2%), EASY 384 (20.2%). All exceed ≥ 30 obs. PIVOTING = 27 (1.4%).
Is any state degenerate?	Mostly yes	No state >80% dominance. PIVOTING statistically thin.
Can we ship v2 fed_cycle to production?	No	Shadow labels exist; production still uses legacy 7-state.

Doubt for Rohit sir: PIVOTING has only 27 Fridays. Merge into EASING, widen the definition, or add PAUSING as a fifth state? Fed is currently on hold with no v2 label (job tracker T-01).

A2: curve_regime (4 states + fiscal caveat)

Question	Answered?	Answer
Does fiscal deficit >5% weaken inversion signal?	Directionally yes	13 inverted episodes. No-offset (deficit $\leq 5\%$): n=12, 41.7% bearish 3m. Fiscal-offset: n=1, 0% bearish (likely 2022–23).
Are 4 curve states stable in backfill?	Yes	All four appear. Oct 2022: 14 inverted weeks via F2 rule.

Doubt for Rohit sir: Fiscal-offset bucket has n=1. Cannot confirm 2022–23 tagging statistically.

A3: val_regime + CAPE velocity

Question	Answered?	Answer
Does CAPE velocity add signal beyond static level?	Not yet	Level extreme 3m: n=863, 74.2% SPX up, avg +3.08%. Velocity rank delta 6m: n=531, 74.6% up, avg +2.68%. Hit rates differ 0.4 pp only.
Does triple storage help Combo E?	Partially	Combo E high-CAPE 6m: n=507, 79.1% hit, avg +6.41%. Moderate-CAPE bucket: n=0.

Doubt for Rohit sir: Fresh cross into EXTREME CAPE: n=0 with current definition. Cannot test "fresh crossing vs sitting at extreme for 3 years" yet.

A4: geo_overlay (6 → 3 states)

Question	Answered?	Answer
Is 3-state geo more reproducible than 6-state?	Yes (qualitatively)	NEUTRAL 1,855 (97.6%), ELEVATED_RISK 25 (1.3%), CRISIS 21 (1.1%).
Does geo slice combo performance meaningfully?	No	FM geo slices mostly n<10. CRISIS n=2, ELEVATED_RISK n=1 at extreme-short FM.

A5: liquidity (2 → 4/9 states)

Rohit sir's plan moves liquidity from binary GLOBAL_EASY / GLOBAL_TIGHT to a **2×2 grid**: easy vs tight **level** (from NFCI) crossed with improving vs tightening **direction** (from WALCL month-over-month change). That is the right design. What the shadow backfill showed is that real data does not always land in one of four clean quadrants.

How the 9 states are built (regime_v2_shadow.py):

Input	Rule	Output
$NFCI \leq -0.3$	Easy financial conditions	Level = EASY
$NFCI \geq +0.3$	Tight financial conditions	Level = TIGHT
Between -0.3 and $+0.3$	Neither extreme	Level = NEUTRAL (not in plan's 2×2)
$WALCL\ MoM > +0.3\%$	Balance sheet expanding	Direction = IMPROVING
$WALCL\ MoM < -0.3\%$	Balance sheet shrinking (QT)	Direction = TIGHTENING

Input	Rule	Output
Between -0.3% and $+0.3\%$	No clear move	Direction = FLAT (not in plan's 2×2)

Label format is `{LEVEL}_{DIRECTION}`, giving **3 levels × 3 directions = 9 states**. The plan's pure 4-state grid would require every Friday to be EASY or TIGHT (no NEUTRAL) and IMPROVING or TIGHTENING (no FLAT).

Full backfill distribution (1,901 Fridays):

State	Count	% of sample	Passes ≥30 obs?
EASY_FLAT	746	39.2%	Yes
EASY_IMPROVING	403	21.2%	Yes
EASY_TIGHTENING	287	15.1%	Yes
NEUTRAL_FLAT	219	11.5%	Yes
NEUTRAL_TIGHTENING	72	3.8%	Yes
NEUTRAL_IMPROVING	62	3.3%	Yes
TIGHT_IMPROVING	50	2.6%	Yes
TIGHT_TIGHTENING	32	1.7%	Yes
TIGHT_FLAT	30	1.6%	Yes

Worth noting: **50.8%** of all Fridays sit in a `*_FLAT` direction bucket (965/1,901), and **26.4%** are `NEUTRAL_*` level (501/1,901). That is why the backfill produced 9 labels, not 4. Forcing a pure 2×2 would misclassify roughly half the history whenever WALCL MoM is near zero (common during QT pause, post-QE plateau, or noisy weekly prints).

Does WALCL direction add signal? I sliced FM positioning events by liquidity_v2 at the 3m horizon:

Band	Liquidity slice	n	SPX up 3m	Notes
Extreme short FM (<15th)	EASY_FLAT	6	50.0%	No clear edge
Extreme short FM	EASY_IMPROVING	10	60.0%	Similar to FLAT

Band	Liquidity slice	n	SPX up 3m	Notes
Extreme short FM	EASY_TIGHTENING	10	50.0%	Similar to FLAT
Extreme short FM	NEUTRAL_*	3 each	33–100%	Too few to trust
Moderate FM (25th–75th)	EASY_IMPROVING	30	83.3%	Highest slice
Moderate FM	EASY_FLAT	20	70.0%	
Moderate FM	EASY_TIGHTENING	23	65.2%	~18 pp below IMPROVING
Moderate FM	TIGHT_*	1–2	n/a	Unusable

Direction is **encoded correctly** in the labels (EASY_TIGHTENING vs EASY_IMPROVING are distinct periods). But at the FM-event level, hit rates within the EASY level cluster around 50–60% for extreme short and 65–83% for moderate. The spread is not large enough yet to treat liquidity direction as a standalone combo filter. TIGHT_* buckets are too thin (n=30–50 in the full backfill, n=1–2 at FM events) for any regime-conditional conclusion.

Question	Answered?	Answer
Does WALCL direction distinguish tightening vs improving?	Built yes; signal unproven	Labels separate IMPROVING/TIGHTENING/FLAT using WALCL MoM $\pm 0.3\%$ thresholds. Distribution shows direction matters descriptively (EASY_IMPROVING 403 vs EASY_TIGHTENING 287 Fridays). FM slices do not show a reliable performance gap at event level.
Is 4-state 2×2 enough or do we need FLAT variants?	Yes: keep 9 for labels, collapse to 4 for analytics	See recommendation below.

My answer on 4 vs 9 states (from backfill + logic):

I recommend a **two-tier approach**, not picking one number for everything:

1. **Production regime storage and classifier output: use 9 states.** Forcing 4 would mislabel ~50% of Fridays where WALCL direction is flat or NFCI is in the neutral band. The 9-state scheme is honest to the data and matches how `liquidity_v2()` is already implemented. Every cell passes the ≥ 30 obs rule in the full backfill (thinnest is TIGHT_FLAT at n=30).

2. **Combo hit-rate tables, beta filter, and briefing footnotes: collapse to 4 pure 2×2 buckets** when slicing performance, because 9-way slices are too thin at the event level (FM extreme short: n=6–10 per EASY cell; TIGHT cells nearly empty).

Collapse rules I would use for analytics:

9-state label	Collapsed 4-state bucket
EASY_IMPROVING	EASY + IMPROVING
EASY_TIGHTENING	EASY + TIGHTENING
EASY_FLAT	EASY + IMPROVING if prior 4wk WALCL trend positive, else EASY + TIGHTENING (or hold FLAT as "no direction call")
NEUTRAL_IMPROVING	EASY + IMPROVING (NFCI < 0 → lean easy)
NEUTRAL_TIGHTENING	EASY + TIGHTENING (or TIGHT + TIGHTENING if NFCI > 0)
NEUTRAL_FLAT	NEUTRAL level: split by NFCI sign or exclude from 4-way slice
TIGHT_IMPROVING	TIGHT + IMPROVING
TIGHT_TIGHTENING	TIGHT + TIGHTENING
TIGHT_FLAT	TIGHT + dominant recent WALCL trend

1. **Do not drop FLAT or NEUTRAL from storage.** FLAT is economically real (QT on hold, balance sheet plateau, weekly WALCL noise). NEUTRAL NFCI is real (mildly loose conditions that do not clear the ± 0.3 easy/tight gate). Dropping them would recreate the old binary GLOBAL_EASY/TIGHT problem under a new name.

Remaining doubt for Rohit sir:

- What is the final decision on the 4 vs 9 states?
- The collapse rules for NEUTRAL_FLAT and EASY_FLAT are judgment calls. Do you prefer NEUTRAL level folded into EASY (majority of NEUTRAL Fridays have NFCI slightly negative) or kept as a third level in the classifier prompt only?

Part B: 14th variable and history windows

B1: TWY_ROC (#14)

Question	Answered?	Answer
Did TWY_ROC call Apr 2025 bottom before lagging fed labels?	Yes	Apr 7 2025: TWY_ROC -0.55 pp DOVISH (DGS2 3.73%). Legacy fed still TIGHTENING/PAUSING.
Are ± 0.30 pp bands validated?	Partially	Anchor passes (well below -0.30). No full historical band sweep.
Is TWY_ROC excluded from combos?	Yes	298 signatures from 13 vars only. 13,089 generic fires without TWY_ROC leg.

B2: Dual percentile storage

Question	Answered?	Answer
Does dual percentile storage work?	Yes	14,457 rows with both unconditional + regime pctile. 0 unconditional-only rows.
Does <50 fallback logic work when needed?	Not tested	0 fallbacks in backfill. Every fed_cycle had ≥ 50 obs in practice.
Are history windows correct per variable?	No	4 FAIL: HY/VIX/VXTS configured <code>full</code> (plan wants <code>rolling_3y</code>); WALCL was <code>rolling_3y</code> (plan wants <code>full</code>). WALCL fixed in production nightly 2026-06-09 but B4 audit not re-run.

B3: Triple CAPE storage

Question	Answered?	Answer
Which CAPE storage combo predicts best?	Preliminary: level	Level wins avg return by $+0.40$ pp. Not a rigorous multivariate test.
Does velocity beat level for Combo E?	No clear win	High-CAPE Combo E 6m strong regardless of velocity tier.

Deliverable B summary

TWY anchor **PASS**. Dual storage **PASS**. Window audit **FAIL** (4 vars). TWY not in nightly pull yet.

Part C: Emission probability vectors

Question	Answered?	Answer
Can we store 14 daily percentile vectors?	Yes	8,805 daily rows backfilled.
Do sub-threshold readings accumulate useful signal?	Maybe	VIX 65th–79th pctile (below RARE): n=7, 85.7% positive 3m. Too small for statistical gate.
Do vectors detect shifts earlier than binary?	No (so far)	864 VIX RARE events: median lag binary vs vector = 0.0 days.
When can HMM training start?	Not yet	Need 6+ months live vectors. Clock at 0 months.

Doubt for Rohit sir: Prototype HMM did not improve Combo B (–1.2 pp) or D (–1.9 pp). Is ~Dec 2026 still the right HMM target?

Part D: HMM layer

Question	Answered?	Answer
Can HMM posteriors feed classifier as soft prior?	Not in production	Prototype: Jun 2026 sample Risk-On 42.9%, Risk-Off 23.8%, Transition 33.3%.
Does HMM detect shifts before binary thresholds?	Not validated	D3 shift-timing backtest not completed.
HSMM for combo duration exit probability?	Not started	Correctly deferred.
Does HMM improve Sharpe/win rate/drawdown?	No on win rate	Combo B: 79.8% → 78.6% with Risk-Off filter (n=56). Combo D: 28.1% → 26.2% (n=103). Sharpe/drawdown not reported.
When is production HMM ready?	~Dec 2026 earliest	Live C1 not wired.

Part E: Cancel probability

Question	Answered?	Answer
Can we compute Combo C cancel prob via MC?	Yes	10,000 GBM paths. WTI leg 8.31%, CPI leg 27.04%, combined 2.25%.
Is probability calibrated to history?	No	4 historical episodes, 0 realized cancels vs 2.25% predicted.
Live cancel % on briefing?	Not yet	Function built, not displayed.
Combo D/F/G cancel formulas?	Documented only	Only Combo C implemented as reusable function.

Part F: Formal regime definitions

F1: TIGHTENING-LATE

Question	Answered?	Answer
Does F1 match Oct 2022?	No	tightening_late_f1 = False. FFR ~3.25% below >3.5% threshold. Legacy was HIKING_LATE. hiking_period_f3 = true.
All fed states have numeric rules?	No	F1 proposed for TIGHTENING-LATE only. Full 4-state quant defs not validated.

F2: INVERTED

Question	Answered?	Answer
Reproducible INVERTED from T10Y2Y?	Yes (shadow)	T10Y2Y < 0 for ≥ 4 consecutive weeks. Oct 2022: 14 inverted weeks.

F2a: STEEPENING

Question	Answered?	Answer
STEEPENING detectable from numeric rules?	Yes (shadow)	$\geq +15$ bps/4wk RARE, $\geq +40$ EXTREME.

F4: Steepening-of-inversion short grid

Trough	Steepen 4wk	n	SPX down 3m
−50 bps	+15 bps	17	17.6%
−50 bps	+40 bps	4	25.0%
−80 bps	+15 bps	9	33.3%
−80 bps	+40 bps	2	0%

Question	Answered?	Answer
−50 vs −80 trough?	Grid run, no stat winner	−80/+15 best cell (33.3%, n=9) but below 55% bar.
+15 vs +40 steepening?	+15 wins on n	+40 cells n=2–4, meaningless.
Can F4 be promoted on backtest alone?	No	Best 33.3% vs ≥80% naming gate. Mechanism+analog only.

Part G: Persistence signals

G1: SEVEN_WEEK_GRIND

Question	Answered?	Answer
Is 7-week grind a standalone short?	No	n=2 episodes. Both negative 6m (avg −5.91%). standalone_short_ok = false. Matches plan (Combo E amplifier).
Wire as Combo E amplifier?	Built, not live	persistence_fires table in shadow. Briefing does not show grind status.

G2: VIX_SUPPRESSED

Question	Answered?	Answer
Does VIX suppressed precede vol spike ~50%?	No	n=1,973 periods. Lead rate to VIX>25 within 35d: 8.5% (168/1,973). Plan claimed ~50%.
Precursor for Combo D?		

Question	Answered?	Answer
	Directionally yes, weakly	~91.5% false watches if treated as sell trigger. Correct framing: watch flag only.

Part H: Nine-step combo discovery pipeline

Step	Question	Answered?	Answer
1 Detection	298 combos scanned?	Yes	225 with ≥ 1 fire. 13,089 total generic fires.
2 Forward returns	SPX 1m–12m stored?	Yes	Complete in pipeline JSON.
3 Regime tagging	v2 labels used?	Partially	Legacy regime tags on existing combo_fires.
4 Surfacing	≥ 3 fires, $\geq 60\%$ HR?	Yes	187 surfaced.
5 Beta filter	Beat hostile regimes?	Yes	132 pass ($\geq 55\%$ and $\geq 60\%$ both reported).
6 Directionality	≥ 2 of 5 dimensions?	Yes	132 pass (same set as beta).
7 Tavila story	Economic narrative?	No	use_claude=False. Skipped.
8 Naming gate	≥ 5 fires, $\geq 80\%$ HR?	Partially	62 promotion candidates. 0 promoted to new named combos.
9 Output table	Cancel prob attached?	No	Part E not wired to live output per combo.

Pipeline funnel:

Stage	Count
Signatures	298
With fires	225

Stage	Count
Surfaced	187
Beta + directionality pass	132
Promotion candidates	62
Promoted to production names	0

Doubts for Rohit sir: Beta 55% or 60% for 62 candidates? Re-tag with v2 regimes before final review? Run Tavily step 7 before any promotion?

Part I: Sample-size (covered in brief)

Rule	Threshold	Observed in run
Regime-conditional percentile minimum	≥ 30 obs	PIVOTING fails (n=27)
Fallback to unconditional	< 50 obs	0 fallbacks triggered
Statistical gate	≥ 5 fires	Applied to FM bands, unnamed combos
Mechanism gate	2–4 fires OK	F4, Combo B washout

Two evidence standards are working as designed: F4 correctly stays on mechanism+analog because win rates are too low for statistical promotion.

FM and regime isolation (Rohit sir's additional ask on WhatsApp)

Extreme short FM (<15th pctlile)

Test	n	SPX up 3m	FM wrong / contrary
Combo B fires (incl. WATCH)	89	79.8%	Validates direction vs Rohit sir's 87.5% (7/8)
Raw FM <15 crossings	35	60.0%	Weaker than headline claim
EASY fed_cycle slice	6	83.3%	Strongest regime
FLAT curve slice	6	33.3%	Signal breaks here

Question	Answered?	Answer
Is extreme-short FM a contrary indicator?	Conditionally yes	Works best with full Combo B legs, not FM alone.
Why 89 vs "8 confirmed"?	Explained	DB counts WATCH rows with partial legs. Confirmed-only slice not yet run (open P0 task).

Extreme long FM (>85th pctile)

Horizon	Raw FM: SPX down	Raw FM: wrong (SPX up)	Combo D: wrong
1 week	41.0%	59.0%	61.5%
3 months	17.9%	82.1%	71.9%

Question	Answered?	Answer
FM wrong 72–85% at short horizons?	Partially	Backtest shows ~59–62%, roughly 10–20 pp below Rohit sir's band.
Signal degrades at 3m?	Yes (correction doesn't happen)	82% FM wrong at 3m for raw band.
Regime impact on Combo D?	Yes	HIKING_LATE 18.3% SPX down 3m (n=197) vs CUTTING_LATE 43.2% (n=155).

Moderate FM (25th–75th)

Metric	Value
Crossings	84
SPX up 3m	76.2%
Avg 3m return	+3.15%

Rohit sir was right to be skeptical. This looks like equity drift, not an independent FM edge. No alpha from fading or following moderate FM.

Named combos A–G by fed cycle (3m)

Combo	Dir	n	Overall hit	Best slice	Worst slice
A	Bear	174	23% down	CUTTING_LATE 50% (n=26)	QE 20% (n=112)
B	Bull	89	79.8% up	HIKING_LATE 83% (n=48)	CUTTING_LATE 76% (n=41)
C	Bull	4	0% up	n too small	n/a
D	Bear	452	28% down	CUTTING_LATE 43%	HIKING_LATE 18%
E	Bear	507	20% down	Flat ~14–27%	QE 14%
F	Bull	704	74.9% up	QE 82% (n=212)	CUTTING_LATE 64%
G	n/a	0	No fires	n/a	n/a

Question	Answered?	Answer
Does regime materially change combo performance?	Yes	D: 2.4× spread HIKING vs CUTTING. F: 18 pp QE vs CUTTING spread.
Full 5-dimension slicing done?	Partially	fed_cycle reliable; other dims mostly n<10.
Combo C working?	No in sample	4 fires, 0% 3m up, avg +17.8% (market rose against signal).

My doubts and questions

- PIVOTING at n=27: is it a real state or label noise? Merging into EASING is pragmatic but might hide genuine pivot weeks.
- I have not stripped unconditional market drift from the moderate FM 76.2% result. Does it still look special vs buy-and-hold?

- Combo C has $n=4$ and 0% hit. Should we show C hit rates in the briefing at all until we have more completed episodes?
- VIX suppressed lead rate is 8.5% vs plan's ~50%. Did we measure the wrong window, or is the plan figure an informal estimate that needs updating?
- For production GO-live, what is the rollback plan if v2 fed_cycle labels confuse readers who have used legacy names for months?
- SSI short gate is weak at 26% SPX-down when $SSI \geq 0.85$. Is that acceptable for live sizing, or do we need a tighter gate despite the backtest?